

Evidence PDF

Introduction to Programming 2 (Python) - Assignment 3

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Group	SE 2505
File	main.py
Contains	Tasks 1-10, console execution, generated files

Generated files included in the archive:

- students.txt
- employees.csv
- employees_updated.csv
- ems.csv
- evidence_console_output.txt
- main.py

Task 1

```
--- Task 1 ---
Name: Alice
Age: 25
City: Astana
Name: Bob
Age: 30
City: Almaty
Enter name: Enter age: Enter a hobby: Name: Charlie
Age: 22
City: Astana
Hobby: Chess
```

Task 2

```
--- Task 2 ---
Enter a list of numbers (separated by spaces): Enter threshold X: Original list: [1, 2, 3, 4, 5, 12, 13]
Squared numbers: [1, 4, 9, 16, 25, 144, 169, 196]
Even numbers: [2, 4, 12, 14]
Numbers greater than 10: [12, 13, 14]
```

Task 3

```
--- Task 3 ---
Enter a new book details:
Title: Author: Number of pages: The book 'To Kill a Mockingbird' by Harper Lee has 281 pages.
The book 'The Great Gatsby' by F. Scott Fitzgerald has 180 pages.
The book '1984' by George Orwell has 328 pages.
```

Task 4

```
--- Task 4 ---
Enter rectangle width: Enter rectangle height: Rectangle(width=5.0, height=10.0)
Area: 50.0
Perimeter: 30.0
```

Task 5

--- Task 5 ---

Car Make: Car Model: Year: Number of doors: Vehicle: Honda Civic (2022)

Doors: 4

Beep beep!

Task 6

--- Task 6 ---

Enter radius for the circle: Enter side length for the square: This is a circle with radius 5.0.

Area: 78.54

This is a square with side 4.0.

Area: 16.0

Task 7

--- Task 7 ---

Author Name: Author Nationality: Book Title: Publication Year: Book: "The Old Man and the Sea" (1952)

Author: Ernest Hemingway (American)

Task 8

--- Task 8 ---

Enter new student name: Enter student ID: Enter three grades (spaces): Data written to students.txt.

--- Loaded Students ---

Alice (ID: 001) - Average: 89.0

Bob (ID: 002) - Average: 75.0

Charlie (ID: 003) - Average: 92.3

Task 9

--- Task 9 ---

New Employee Name: Age: Department: Salary: Written 3 employees to employees.csv

Applied 5% raise to all employees.

Updated data written to employees_updated.csv

Task 10

--- Task 10 ---

1. Add Employee
2. Save to File
3. Load from File
4. Payroll Report
5. Exit

Choose option: Type (1:Hourly, 2:Salaried, 3:Commission): Name: ID: Annual Salary:

1. Add Employee
2. Save to File
3. Load from File
4. Payroll Report
5. Exit

Choose option:

Payroll Report:

John Doe (SalariedEmployee) - Monthly Pay: \$5000.00

1. Add Employee
2. Save to File
3. Load from File
4. Payroll Report
5. Exit

Choose option: Saved.

1. Add Employee
2. Save to File
3. Load from File
4. Payroll Report
5. Exit

Choose option: Loaded.

1. Add Employee
2. Save to File
3. Load from File
4. Payroll Report
5. Exit

Choose option:

Payroll Report:

John Doe (SalariedEmployee) - Monthly Pay: \$5000.00

1. Add Employee
2. Save to File
3. Load from File
4. Payroll Report
5. Exit

Choose option:

File I/O Proof

students.txt

```
Alice|001|85,90,92  
Bob|002|70,80,75  
Charlie|003|95,90,92
```

employees.csv

```
name,age,department,salary  
Alice,25,HR,50000.0  
Bob,30,Dev,60000.0  
Mark,35,IT,75000.0
```

employees_updated.csv

```
name,age,department,salary  
Alice,25,HR,52500.0  
Bob,30,Dev,63000.0  
Mark,35,IT,78750.0
```

ems.csv

```
Salaried,John Doe,E001,60000.0
```